

Technology Stack

Date	29 June 2026
Team ID	N/A
Project Name	OptiCrop Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Define Your Technology Stack: Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML, CSS, JavaScript, Bootstrap.	Creates a responsive, user-friendly interface for farmers.

2	Backend / Server-Side	Python (Flask)	Handles user requests, AI prediction logic, and server-side processing.
3	Database / Data Storage	MySQL	Stores user details, crop data, and prediction records securely.
4	Cloud / Hosting / Deployment	Render or Heroku	Provides easy deployment and online access to the application.

5	Version Control & CI/CD	Git &	Manages source code, collaboration. &
		GitHub	version tracking.
6	Third-Party APIs / Other Tools	OpenWeatherMap API, Scikit-learn, Pandas, NumPy	Weather API provides real-time weather data; Scikit-learn is used for AI crop prediction; Pandas and NumPy are used for data processing.